

Interactive Decision-Making Integrating Graph Neural Networks and Model Predictive Control for Autonomous Driving

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Abstract—Driving on public roads is inherently an interactive task, *i.e.*, autonomous vehicles' (AVs) actions will influence nearby traffic participants' reactions, and vice versa. Decision-making for AVs in highly interactive driving scenarios (*e.g.*, dense traffic) requires accurately forecasting the impact of the AVs' intention on nearby traffic participants' motion. To this end, a hierarchical decision-making framework (HDM) is proposed to navigate through interactive scenarios safely and efficiently. Specifically, the upper layer of the HDM serves as a coarse-level policy generator, which utilizes the plan-informed graph attention network (P-GAT) to provide interaction-aware guidance. The P-GAT predictor takes the historical states of nearby traffic participants, road structure information, and AVs' potential intentions as inputs. Subsequently, it predicts the motions of other traffic participants in response to potential actions of the AVs, which is then systematically evaluated to generate interactive guidance. Furthermore, the learned policy is utilized to guide the lower layer, which utilizes a fine-level model predictive control (MPC)-based planner to ensure safety and kinematic feasibility. Finally, to validate the effectiveness of HDM, both qualitative and quantitative experiments are carried out. More importantly, the hardware-in-the-loop (HiL) experiment is also implemented, including mandatory lane change in dense traffic flow and interaction with the human driver. The results demonstrate that the proposed HDM can improve driving safety and efficiency compared with baselines.

Index Terms—Autonomous driving, interactive decision-making, graph neural network, model predictive control.

I. INTRODUCTION

AUTONOMOUS driving technology has attracted widespread attention due to its immense potential to improve traffic efficiency, enhance road safety, and reduce energy consumption [1], [2], [3]. Decision-making is one of the central component, *i.e.*, the core manifestation of the intelligence of autonomous vehicles (AVs) [4]. It requires safe

handling of uncertainty risks in external driving environments and internal autonomous driving systems [5], [6], [7]. Moreover, AVs ought to possess the ability to safely interact with other traffic participants—a fundamental skill that human driver instinctively possess, yet remains unresolved.

Specifically, when deployed on public roads, AVs need to share the driving space with other uncontrolled traffic participants including human-driven vehicles and pedestrians, which poses a non-trivial challenge for the decision-making task [8]. In particular, navigating through real-world dense traffic scenarios that requires the AV to cooperate with nearby traffic participants, such as lane merging and unprotected left turns, remains a significant hurdle [9]. Currently, most classical AV stacks assume a one-way interaction, *i.e.*, only the AV is influenced by other traffic participants. However, this assumption is not valid since the AV also influences other traffic participants' decisions in reality. Decisions that only consider one-way interaction, without considering the mutual influence could contribute to overly conservative plans of the AV or “freezing robot problem”. To solve this issue, the key lies in accurately predicting how the AV's action affects surrounding traffic participants' reactions, and utilizing their cooperativeness to accomplish the driving task safely and efficiently. Accordingly, this paper proposes a plan-informed graph attention network (P-GAT) to better capture the bi-directional interaction between the AV and nearby traffic participants. In addition, P-GAT also considers the interaction features between the AV and road structures, as well as the interaction features among surrounding traffic participants. These enable P-GAT accurately forecast the impact of AV's intention on surrounding traffic participants' motion. Subsequently, an interaction-aware decision-making framework is developed, which integrates P-GAT and optimization-based planner in a hierarchical manner, named hierarchical decision-making (HDM). The upper layer of HDM, *i.e.*, coarse-level policy generator is developed based on P-GAT, which provides interaction-aware guidance for the fine-level model predictive control (MPC)-based planner. It is worth noting that, inspired by [10], unlike existing hierarchical decision-making methods [11], the P-GAT model is not trained offline using datasets. Instead, P-GAT is trained online by collecting data through the interaction between the HDM and the traffic environment. As a result, P-GAT is better able to capture the impact of the AV's intention on surrounding traffic participants.

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The main contributions of this paper are summarized as follows:

1) A plan-informed prediction network, *i.e.*, P-GAT is proposed to capture the bi-directional interaction between the AV and nearby traffic participants. P-GAT can forecast interactive motions of nearby traffic participants in response to the AV's potential plans.

2) An interactive decision-making method, *i.e.*, HDM is proposed. The upper layer serves as a coarse-level policy generator based on P-GAT to provide interaction-aware guidance. Then, the lower layer, *i.e.*, a fine-level MPC-based planner incorporates the interaction-aware guidance to generate safe and kinematic-feasible trajectories.

3) Unlike existing prediction-then-planning hierarchical methods, where the prediction model is mostly pre-trained offline, P-GAT is trained by collecting data through HDM interacting with the traffic environment, similar to the reinforcement learning training manner.

The rest of this paper is organized as follows. Section II presents the related work. The overall structure of the proposed decision-making framework is depicted in Section III. Section IV elaborates on the interactive trajectory prediction model P-GAT. The coarse-level decision-making policy generator and fine-level MPC-based planner are presented in Section V. Section VI presents the results and discussion, followed by the conclusion in Section VII.

II. RELATED WORK

Recent studies have underscored the critical need to enhance the interaction awareness of AVs, recognizing that efficient capture of inter-vehicle interactions is paramount [12], [13], [14]. This awareness is a fundamental prerequisite for safe and effective decision-making. Currently, interaction-aware decision-making approaches can be broadly categorized into several types: game theory-based methods, partially observable Markov decision process (POMDP)-based methods, learning-based methods, and joint prediction and planning methods [15].

Game theory provides an effective paradigm for addressing interactive decision problems, encompassing various strategies such as Stackelberg games, Grand Coalition games, k-level games, *etc.* For example, Hang et al. proposed a human-like decision-making framework where Nash equilibrium and Stackelberg were applied to non-cooperative decision-making, while accounting for the driving style of AVs [16]. However, solving these games presents two primary challenges: firstly, the reward functions of surrounding traffic participants are usually unknown, and secondly, determining a Nash equilibrium or other equilibrium of the game can be computationally demanding [17]. POMDP is another widely utilized technique for interactive decision-making. It excels particularly in modeling uncertainty in traffic participant intentions and accounting for environmental occlusion [18]. To consider the interaction, Hubmann et al. formulated decision-making problems as POMDPs by expanding the state space to include neighboring vehicles, thus enabling interactive behavior [19]. In addition, an interaction-aware POMDP planner was proposed for autonomous driving in dense urban environments [20].

It employed multi-step occupancy grid maps to represent the states of surrounding traffic participants, including their uncertain intentions. Similar to game theory, as the number of traffic participants increases, particularly in dense traffic scenarios, computational burden remains a significant challenge.

Learning-based techniques, particularly deep reinforcement learning (DRL), have been increasingly employed to devise interaction-aware decision-making policies in complex or dense traffic scenarios [21], [22], [23]. For instance, Saxena et al. utilized reinforcement learning to implicitly model interactions between AVs and other traffic participants, exploiting the cooperativeness of drivers to navigate dense traffic by opening gaps [24]. Moreover, [25] employed the soft actor-critic (SAC) algorithm to learn a policy maximizing long-term rewards by leveraging interaction effects, guiding a local optimization planner through dense traffic scenarios. Similarly, a hierarchical reinforcement learning-based decision-making framework was proposed to learn interactive left-turn policies at unsignalized intersections [26]. More recently, the scene representation learning technique is commonly combined with DRL for highly interactive driving scenarios. For instance, SPformer was proposed [27], which designed the transformer encoder based on ViT to extract the interactive information between agents, thereby speeding up the learning process of the DRL algorithm and improving the quality of the learned policy. Moreover, [28] introduced an AV-centric spatiotemporal attention encoding (STAE) mechanism for learning the dynamic interactions with different surrounding vehicles. STAE provided a comprehensive state representation for the SAC-based decision-making module. In addition, Liu et al. [29] constructed a multistage Transformer (MST) encoder to model not only the interaction between the AV and its neighbors but also intention awareness between the agents and their candidate routes. Overall, DRL empowered by deep-structured neural networks, has demonstrated superiority in making interaction-aware decisions. However, practical applications of DRL are constrained by safety, reliability, and interpretability concerns.

Recent works suggest that integrating neural network-based predictor with planner in an interdependent joint step can lead to safe, efficient and interaction-aware driving policies [15]. Bae et al. introduced a cooperation-aware lane change control framework based on MPC with recurrent neural network, namely, Social GAN [11]. Similarly, another cooperation-aware method for determining the appropriate time instant and maneuver in lane changes was proposed [30], which utilized the graph convolution network to predict surrounding vehicles' interactive behavior with incorporating AV's decision. In these methods, the predictor took only the historical trajectories of all traffic participants as input and did not directly encode the AV's intentions. Therefore, when evaluating candidate control commands for the AV, the predictor outputted predictions only for the next time step. It then considered the influence of the AV's actions on surrounding traffic participants in an iterative manner, potentially increasing computational burden. Moreover, an interaction-aware trajectory planner utilizing a differentiable neural network-based predictor was proposed for dense lane change scenarios, which analytically integrated

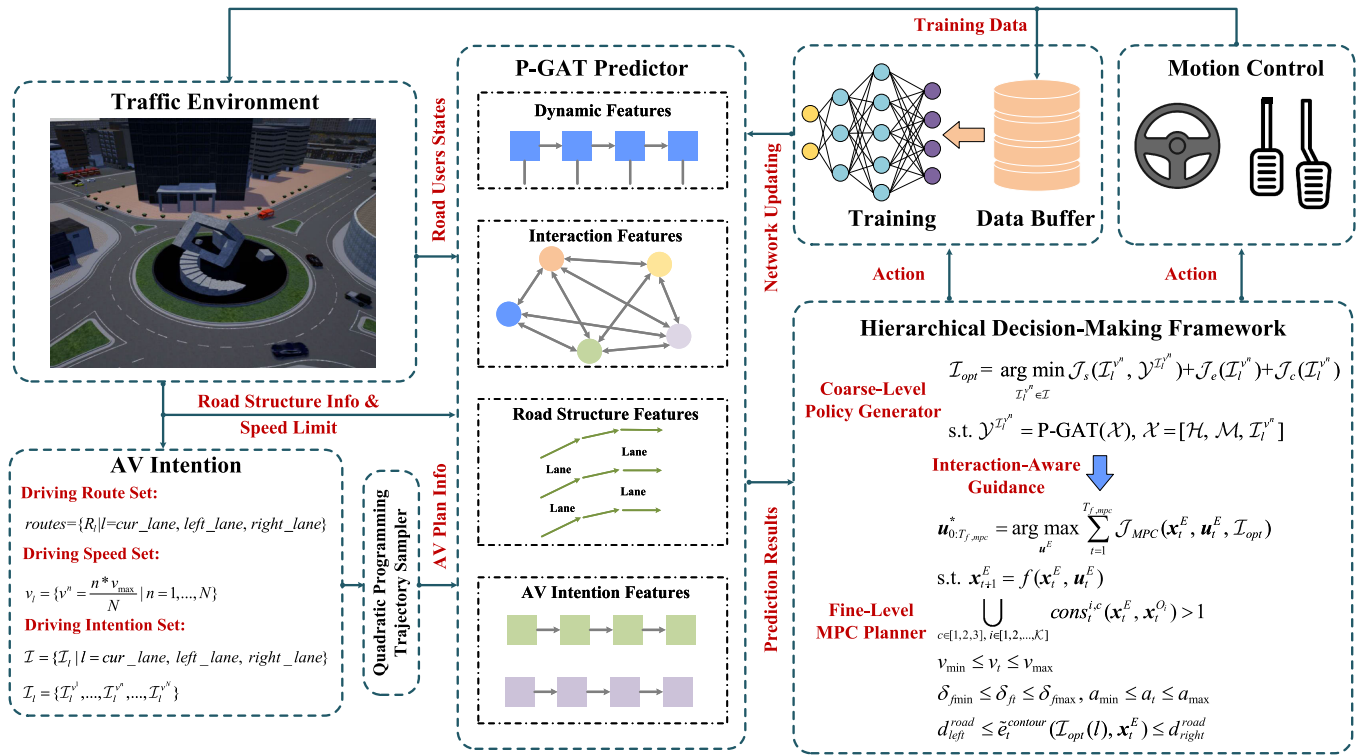


Fig. 1. The overall structure of the proposed hierarchical decision-making method.

predictor with MPC [31]. Other notable works include DiffStack [32], DIPP [33], *etc.*, and the core idea of these methods is to integrate neural network-based prediction models with differentiable optimization-based planner. Nevertheless, the computational burden due to the complexity remains a limitation. More recently, an interaction-aware decision-making framework was proposed, employing Transformers to effectively encode the driving scene and incorporate the AV's future plan in decoding the predicted trajectories [10]. However, this approach did not account for vehicle kinematic constraints, and the safety and efficiency were largely dependent on the completeness of the candidate trajectories of the AV, which was closely related to computational burden.

III. OVERALL STRUCTURE OF THE HIERARCHICAL DECISION-MAKING FRAMEWORK

A. Framework

Fig. 1 shows the overall structure of the proposed decision-making framework. First, assume that the road structure and speed limits can be obtained from the high-definition map. In addition, the motion states of surrounding traffic participants can be accurately obtained via the AV's onboard sensors. Second, based on the structure of the road and the speed limits, the possible decision making intentions of the AV can be roughly determined. Specifically, lateral decision intentions include changing lanes to the left, changing lanes to the right, and keeping lane. As for longitudinal decision intentions, here, based on speed limit information, they are expressed as discrete desired driving speeds. Note that the determined decision intentions are primarily used to consider the impact of

the AV's actions on surrounding traffic participants. Therefore, longitudinal decision intentions do not need to be excessively discretized. Third, a plan-informed prediction model is established that incorporates the historical states of nearby traffic participants, road structure information, and the decision-making intentions of the AVs as input. To extract the intention features, the quadratic programming technique is applied to produce the specific plan information. Fourth, a coarse-level policy generator based on the established prediction model is proposed, which outputs the interaction-aware guidance. Finally, the policy generator guides the fine-level MPC planner to proactively interact with other traffic participants, while remaining safety via the collision avoidance constraints.

B. Vehicle Model

To depict the vehicle dynamic of the AV, the nonlinear kinematic bicycle model is adopted, *i.e.*,

$$\begin{aligned} \dot{x} &= v \cos(\varphi + \beta) \\ \dot{y} &= v \sin(\varphi + \beta) \\ \dot{\varphi} &= \frac{v}{l_r} \sin(\beta) \\ \dot{v} &= a \\ \beta &= \arctan\left(\frac{l_r}{l_f + l_r} \tan \delta_f\right) \end{aligned} \quad (1)$$

where (x, y) is the coordinate for the center of the AV, β is the slip angle, l_f and l_r are the distance from the front and rear axles of the vehicle to the center of gravity, respectively, v indicates the speed in the vehicle body direction, φ is the heading angle, a and δ_f are the control variables, which represents acceleration and steering angle, respectively.

Denote the AV state variable as $\mathbf{x}^E = [x, y, \varphi, v]$, and the control input as $\mathbf{u}^E = [a, \delta_f]$. For the sake of simplicity, the kinematic vehicle model in (1) can be written as $\mathbf{x}_{t+1}^E = f(\mathbf{x}_t^E, \mathbf{u}_t^E)$ with Euler discretization technique. In addition, for the traffic participant i , the state variable is represented as $\mathbf{x}^{O_i} = [x_i, y_i, \varphi_i, v_i]$.

IV. INTERACTIVE TRAJECTORY PREDICTION BASED ON GRAPH NEURAL NETWORK

A. Construction of Driving Intention Set

Firstly, driving in urban areas takes advantage of the presence of lanes, which can be used as guidelines to limit the number of intentions available to the AV [11]. The lateral decision intentions that the AV can execute can be categorized as: left lane change, right lane change, and keeping lane. The corresponding driving paths can be represented as:

$$routes = \{R_l | l = cur_lane, left_lane, right_lane\} \quad (2)$$

where R_l means the candidate driving path, which in this paper, is represented by the center coordinate of the road, i.e., a sequence of waypoints.

In addition, for the candidate driving path R_l , the target speed can be discretized based on speed limits, which is calculated as follows.

$$v_l = \{v^n = \frac{n * v_{\max}}{N} | n = 1, \dots, N\} \quad (3)$$

where $v_{\max}$ means the speed limits, N represents the number of velocity discretizations for the candidate driving path R_l .

Note that discretizing the target speed based on speed limits in (3) aims to reflect the vehicle's longitudinal driving intentions within the prediction model. Since this is applied to design the coarse-level policy, the intention here is tactical. Thus, N can be adjusted based on the scenario complexity. In addition, the driving intention set $\mathcal{I}$ of the AV can be represented as:

$$\begin{aligned} \mathcal{I} &= \{\mathcal{I}_l | l = cur_lane, left_lane, right_lane\} \\ \mathcal{I}_l &= \{\mathcal{I}_l^{v^1}, \dots, \mathcal{I}_l^{v^n}, \dots, \mathcal{I}_l^{v^N}\} \end{aligned} \quad (4)$$

Furthermore, as aforementioned, the candidate driving intention set $\mathcal{I}$ is utilized to forecast the impact of the AV's action on nearby traffic participants' future motion. To extract AV intention feature for the prediction model, $\mathcal{I}$ should be converted to specific trajectories. For the driving intention $\mathcal{I}_l^{v^n}$, the corresponding candidate driving trajectory information can be precomputed via quadratic programming technique. Specifically, the driving intention $\mathcal{I}_l^{v^n}$ provides the target driving lane $\mathcal{I}_l^{v^n}(l)$ and target driving speed $\mathcal{I}_l^{v^n}(v^n)$ for the AV. To compute the future motion trajectory of the AV within T_f horizon, the objective function can be set as:

$$\begin{aligned} \mathcal{L}(\mathbf{x}_t^E, \mathbf{u}_t^E, \mathcal{I}_l^{v^n}) &= \sum_{t=1}^{T_f} \mathbf{e}_t^\top \mathbf{Q} \mathbf{e}_t + (v_t - \mathcal{I}_l^{v^n}(v^n))^2 \\ s.t. \quad \mathbf{x}_{t+1}^E &= f(\mathbf{x}_t^E, \mathbf{u}_t^E) \\ \mathbf{e}_t &= [\tilde{e}_t^{contour}(\mathcal{I}_l^{v^n}(l)), \tilde{e}_t^{lag}(\mathcal{I}_l^{v^n}(l))]^\top \end{aligned} \quad (5)$$

where $\mathbf{Q}$ is the weight matrix, $\tilde{e}_t^{contour}$, $\tilde{e}_t^{lag}$ represent contour error and lag error when tracking the target driving lane, respectively. For more details, please refer to [34].

By solving (5), the future trajectory $\mathcal{T}_l^{v^n}$ of the AV under driving intention $\mathcal{I}_l^{v^n}$ is obtained (the corresponding set of future trajectory is denoted as $\mathcal{T}$). It can be observed that (5) does not include any safety constraint, which can be transformed into a quadratic programming optimization problem. At present, there are numerous solvers available for solving this kind of issue, such as FORCESPRO [35], with computation efficiency reaching the millisecond level.

B. Graph Neural Network Predictor Incorporating AV's Intention

Accurately understanding traffic participants' behaviors is an essential prerequisite for the AV to make safe and intelligent decisions. However, their behaviors are deeply coupled with factors such as nearby traffic participants, road structure, etc. More importantly, the AV's actions also pose a significant impact on surrounding traffic participants behaviors. Therefore, an interactive prediction model incorporating AV's intention will be proposed.

1) *Input and Output*: the input to the prediction model includes the historical trajectories of traffic participants $\mathcal{H}$, road structure $\mathcal{M}$, and trajectory set $\mathcal{T}$ corresponding to the AV's driving intention $\mathcal{I}$, which is denoted as:

$$\mathcal{X}_t = [\mathcal{H}, \mathcal{M}, \mathcal{T}] \quad (6)$$

Assume that there are a total of $\mathcal{K}$ traffic participants to be considered in total, and $\mathcal{H}$ can be specifically represented as:

$$\mathcal{H}_t = [h^0, \dots, h^i, \dots, h^K] \quad (7)$$

For traffic participant i at time t , the historical trajectory h_t^i can be written as:

$$h_t^i = [s_{t-T_h+1}^i, s_{t-T_h+2}^i, \dots, s_t^i] \quad (8)$$

where T_h means the historical horizon, and s_t^i mainly includes positions and velocities of the traffic participant i at time t .

$\mathcal{M}$ includes the waypoints of the paths that traffic participants may travel along, which can be represented as:

$$\mathcal{M}_t = [m_t^0, \dots, m_t^i, \dots, m_t^K] \quad (9)$$

For traffic participant i , assume that there are $\mathcal{R}$ candidate driving paths at time t ,

$$m_t^i = [\{(x_k, y_k) | k \in [1 : waypoints]\}]_{l \in [1:\mathcal{R}]} \quad (10)$$

where (x, y) is waypoints' coordinates, *waypoints* represents the number of waypoints for the l -th candidate path.

$\mathcal{T}$ is the trajectory set that reflects AVs' driving intention $\mathcal{I}$, which can be denoted as:

$$\begin{aligned} \mathcal{T}_t &= \{\mathcal{T}_l^{v^1}, \dots, \mathcal{T}_l^{v^n}, \dots, \mathcal{T}_l^{v^N} | \\ & \quad l = cur_lane, left_lane, right_lane\} \end{aligned} \quad (11)$$

The output of the prediction model is the $\mathcal{K}$ traffic participants' future trajectories under different AV's driving

intentions. For the driving intention $\mathcal{I}_l^{v^n}$, the output can be denoted as:

$$\mathcal{Y}_t^{\mathcal{I}_l^{v^n}} = [f_t^1, \dots, f_t^i, \dots, f_t^K] \\ f_t^i = [(x_{t+1}^i, y_{t+1}^i), (x_{t+2}^i, y_{t+2}^i), \dots, (x_{t+T_f}^i, y_{t+T_f}^i)] \quad (12)$$

where T_f is the prediction horizon.

2) *Interaction With Nearby Traffic Participants:* As aforementioned, the traffic participant's motion is heavily affected by behaviors of neighbors. Firstly, for traffic participant i , to extract the dynamic feature of h_t^i , the gated recurrent unit (GRU) is utilized:

$$\bar{h}_t^i = \text{GRU}(\text{FC}(h_t^i)) \quad (13)$$

where $\text{FC}(\cdot)$ is made up of fully connected networks and activation functions, $\bar{h}_t^i$ means the extracted dynamic feature of the traffic participant i at time t .

The processed dynamic features of $\mathcal{K}$ traffic participants can be expressed as:

$$\bar{\mathcal{H}} = \{\bar{h}_t^0, \dots, \bar{h}_t^i, \dots, \bar{h}_t^K\} \quad (14)$$

Subsequently, graph neural network is employed to extract the interaction feature among traffic participants. In detail, the GAT network [36], a variant of graph neural network, is utilized to capture interaction features. A directed graph with edge attribute can be represented as $\mathbb{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{\mathcal{V}^0, \dots, \mathcal{V}^i, \dots, \mathcal{V}^K\}$ represents a set with $\mathcal{K}$ traffic participants, and $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ denotes a set of directed edges. For node $\mathcal{V}^i$, its feature can be obtained via (14). In addition, an edge e_{ij} pointing from node j to node i is constructed if traffic participant j is within a predefined distance of traffic participant i . It means that the traffic participant i 's behavior is influenced by the traffic participant j . Specifically, traffic participants located within a certain distance (30m in this paper) from the participant i are considered its neighbors, and the neighbors of the node $\mathcal{V}^i$ are defined as $\mathcal{N}(\mathcal{V}^i)$. Thus, the edge set can be denoted as:

$$\mathcal{E} = \{e_{ij} | j \in \mathcal{N}(\mathcal{V}^i), i = 0, 1, 2, \dots, \mathcal{K}\} \quad (15)$$

where e_{ij} denotes the edge from node $\mathcal{V}^j$ to node $\mathcal{V}^i$.

The edge attribute set is defined as:

$$\mathbf{e}^{attr} = \{e_{ij}^{attr} | i, j = 0, 1, 2, \dots, \mathcal{K}\} \quad (16)$$

For edge attribute e_{ij}^{attr} , which can be calculated as follows.

$$e_{ij}^{attr} = [s_j, \varphi_j] - [s_i, \varphi_i] \quad (17)$$

where s_i includes the positions and velocities of the traffic participant i in global coordination system, and φ_i is the heading angle.

The interaction features between traffic participants processed by the GAT can be represented as:

$$G = \{\bar{g}_t^0, \dots, \bar{g}_t^i, \dots, \bar{g}_t^K\} = \text{GAT}(\bar{\mathcal{H}}, \mathcal{E}, \mathbf{e}^{attr}) \quad (18)$$

3) *Interaction With Road Structure:* Road structure is another important factor that affects traffic participants' motion. To capture the interaction features between the traffic participant and road structure, the Transformer network is utilized [37]. In detail, the positional encoding of map waypoint information $\mathcal{M}$ is conducted, i.e.,

$$\tilde{\mathcal{M}} = \text{PE}(\text{FC}(\mathcal{M})) \quad (19)$$

where $\text{PE}(\cdot)$ is the position encoder in Transformer network.

In addition, the attention mechanism will be used to extract interaction features, i.e.,

$$\tilde{\mathcal{M}} = \{\bar{m}_t^0, \dots, \bar{m}_t^i, \dots, \bar{m}_t^K\} = \text{Attention}(G, \tilde{\mathcal{M}}) \quad (20)$$

where $\text{Attention}(\cdot)$ means the attention mechanism operation, G serves as the Q and $\tilde{\mathcal{M}}$ serves as the K,V in the Transformer network.

4) *Interaction With AV's Intention:* To consider the impact of the AV's intention on different traffic participants, the corresponding AV future trajectory should be rotated into the coordinate system of the predicted traffic participants. Specifically, for the predicted traffic participant i with heading angle φ_i , the future trajectory $\mathcal{I}_l^{v^n}$ of the AV under driving intention $\mathcal{I}_l^{v^n}$ is calculated as follows.

$$\tilde{\mathcal{I}}_l^{v^n} = \mathbf{R}(\mathcal{I}_l^{v^n}) \quad (21)$$

where

$$\mathbf{R} = \begin{bmatrix} \cos\varphi_i & -\sin\varphi_i \\ \sin\varphi_i & \cos\varphi_i \end{bmatrix} \quad (22)$$

Furthermore, the $\tilde{\mathcal{I}}_l^{v^n}$ is processed via the fully connected networks, i.e.,

$$\bar{\mathcal{I}}_l^{v^n} = \text{FC}(\tilde{\mathcal{I}}_l^{v^n}) \quad (23)$$

On the other hand, due to the varying relative positions between the AV and surrounding traffic participants, the influence of AV's plans on the future motion of surrounding traffic participants differs. Therefore, to reflect the influence of AV's plans on different traffic participants, the following equation is employed [10]:

$$score = \text{Sigmoid}(\text{FC}(\text{Relu}(\text{FC}(\bar{\mathcal{H}})))) \quad (24)$$

where $\text{Sigmoid}(\cdot)$, $\text{Relu}(\cdot)$ are the activation functions.

Finally, to obtain the future trajectory of the predicted traffic participant i under the AV's driving intention $\mathcal{I}_l^{v^n}$, the GRU is utilized, i.e.,

$$\mathcal{Y}_t^{\mathcal{I}_l^{v^n}} = \text{GRU}([\bar{h}_t^i || \bar{g}_t^i || \bar{m}_t^i] || \bar{\mathcal{I}}_l^{v^n} * score || \text{FC}(h_t^i)) \quad (25)$$

where $||$ means the concatenation operation.

The average displacement error (ADE) is utilized as the loss function of the proposed prediction model, i.e., P-GAT.

$$\mathcal{Y}_t = \mathbf{P} - \text{GAT}(\mathcal{X}_t) \quad (26)$$

V. INTERACTION-AWARE DECISION-MAKING BASED ON MODEL PREDICTIVE CONTROL

A. Coarse-Level Decision-Making Policy Generator

Based on the established P-GAT model, the AV can predict interactive reactions of other traffic participants in response to potential actions. As previously mentioned, the AV potential driving intentions can be roughly determined according to the road structures and speed limits. In this way, the AV can make interactive decisions to facilitate the driving tasks. Note that the policy generator only considers coarse-level intentions of the AV to predict how its actions might affect other traffic participants. Then, the policy generator provides interactive guidance for the fine-level MPC planner to generate safe and kinematic-feasible trajectories. Specifically, to obtain the optimal interactive guidance, each driving intention $\mathcal{I}_l^{v^n}$ in (4) should be evaluated via (26). Therefore, the following metrics are determined considering driving safety, driving efficiency, etc, which are then combined to construct the evaluation function.

First, for each driving intention $\mathcal{I}_l^{v^n}$, the safety metric is defined as:

$$\mathcal{J}_s(\mathcal{I}_l^{v^n}, \mathcal{Y}^{\mathcal{I}_l^{v^n}}) = \lambda_1 \mathbb{I}_{\text{collision}}(\mathcal{I}_l^{v^n}, \mathcal{Y}^{\mathcal{I}_l^{v^n}}) + \lambda_2 \mathcal{L}_2(\mathcal{I}_l^{v^n}, \mathcal{Y}^{\mathcal{I}_l^{v^n}}) \quad (27)$$

where $\mathbb{I}_{\text{collision}}$ denotes whether the AV will collide with surrounding traffic participants under the driving intention $\mathcal{I}_l^{v^n}$. If the collision occurs, the value is set to 1; otherwise, the value is set to 0. $\mathcal{L}_2$ measures the Euclidean distance between the AV and surrounding traffic participants.

$$\begin{aligned} \mathcal{L}_2(\mathcal{I}_l^{v^n}, \mathcal{Y}^{\mathcal{I}_l^{v^n}}) &= \exp(\tau * d_{\min}^2) \\ d_{\min} &= \arg \min_d \{d \mid d = \sqrt{(\Delta x_t^i)^2 + (\Delta y_t^i)^2}, \\ &\quad \forall t \in [1, T_f]\} \end{aligned} \quad (28)$$

where $\exp(\cdot)$ is the exponential function, $d_{\min}$ represents the minimum distance between the AV and the traffic participant with the prediction horizon, τ means the shape parameter that is negative, $\Delta x_t^i, \Delta y_t^i$ are the distances between the AV and the surrounding traffic participant i at the time t .

Second, the AV should maintain high driving efficiency, i.e.,

$$\mathcal{J}_e(\mathcal{I}_l^{v^n}) = \lambda_3 \frac{1}{T_f} \sum_{t=1}^{T_f} |v_t - v_{\max}| \quad (29)$$

where v_t is the driving speed of the AV, and $v_{\max}$ means the speed limit.

Third, the driving comfort should be considered, i.e.,

$$\mathcal{J}_c(\mathcal{I}_l^{v^n}) = \lambda_4 \frac{1}{T_f} \sum_{t=1}^{T_f} (|\dot{v}_t| + |\ddot{v}_t|) \quad (30)$$

where $\dot{v}_t, \ddot{v}_t$ mean the acceleration and jerk of the AV at the time t , respectively.

Finally, the decision-making process of the coarse-level policy generator can be formulated as:

$$\begin{aligned} \mathcal{I}_{opt} &= \arg \min_{\mathcal{I}_l^{v^n} \in \mathcal{I}} \mathcal{J}_s(\mathcal{I}_l^{v^n}, \mathcal{Y}^{\mathcal{I}_l^{v^n}}) + \mathcal{J}_e(\mathcal{I}_l^{v^n}) + \mathcal{J}_c(\mathcal{I}_l^{v^n}) \\ s.t. \quad \mathcal{Y}^{\mathcal{I}_l^{v^n}} &= \text{P-GAT}(\mathcal{X}), \mathcal{X} = [\mathcal{H}, \mathcal{M}, \mathcal{I}_l^{v^n}] \end{aligned} \quad (31)$$

Therefore, based on (31), interactive guidance can be decided upon. Note that the decision-making here is tactical, considering only the coarse-level intention the AV may take, without detailed consideration of vehicle kinematic constraints and collision avoidance constraints.

B. Fine-Level Model Predictive Control-Based Planner

The coarse-level policy generator provides the interactive guidance $\mathcal{I}_{opt}$, which includes the target driving lane $\mathcal{I}_{opt}(l)$ and target driving speed $\mathcal{I}_{opt}(v)$. Next, a fine-level MPC-based motion planner will be designed to incorporate the interactive guidance $\mathcal{I}_{opt}$. First, the cost function of the MPC planner is formulated as:

$$\begin{aligned} \mathcal{J}_{MPC}(\mathbf{x}_t^E, \mathbf{u}_t^E, \mathcal{I}_{opt}) &= w_1 \|\tilde{\mathbf{e}}_t^{\text{contour}}(\mathcal{I}_{opt}(l), \mathbf{x}_t^E)\|^2 + w_2 \|\tilde{\mathbf{e}}_t^{\text{lag}}(\mathcal{I}_{opt}(l), \mathbf{x}_t^E)\|^2 \\ &\quad + w_3 \|v_t - \mathcal{I}_{opt}(v)\|^2 + w_4 \|a_t\|^2 + w_5 \|\delta_t\|^2 \end{aligned} \quad (32)$$

where w_1, w_2, w_3, w_4, w_5 represent the weight coefficients. The first and second terms represent the cost of tracking the target driving lane. The third term means the cost of tracking the target driving speed, and the last two terms are the cost of control effort.

Second, the trajectory generated by the MPC planner needs to adhere to safety constraints to avoid collisions with surrounding traffic participants. Within the prediction horizon of MPC, the future states of the AV can be rolled out using vehicle kinematic dynamics (1). Simultaneously, it needs to forecast the future motion states of surrounding traffic participants, which can be obtained through (26). To construct collision avoidance constraints, it is necessary to represent the occupied regions of the AV and surrounding traffic participants. Generally, the occupied region of traffic participants can be represented by rectangles, circles, or ellipses. In this paper, three circles ($n_c = 3$) with the same radius r_c are used to approximate the occupied area of the AV, i.e., $\mathcal{A}_{ego}(\mathbf{x}_t^E) = \mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3$. An ellipse with a major axis a_i and minor axis b_i is used to approximate the occupied area $\mathcal{A}_i(\mathbf{x}_t^{O_i})$ of the surrounding traffic participant i . Hence, to avoid collision with the surrounding traffic participant i , the following conditions should be satisfied, i.e., $\mathcal{A}_{ego}(\mathbf{x}_t^E) \cap \mathcal{A}_i(\mathbf{x}_t^{O_i}) = \emptyset$. Based on the geometric relationship that ellipses and circles do not intersect, the collision avoidance constraints can be formulated as:

$$\begin{aligned} &\bigcup_{\substack{c \in [1, 2, 3] \\ i \in [1, 2, \dots, \mathcal{K}]}} \text{cons}_t^{i,c}(\mathbf{x}_t^E, \mathbf{x}_t^{O_i}) \\ &= \Delta \mathbf{x}_t^{i,c \top} \mathbf{R}(\varphi_t^i)^\top \frac{1}{\alpha^2} \mathbf{R}(\varphi_t^i) \Delta \mathbf{x}_t^{i,c} \\ &\quad + \Delta \mathbf{y}_t^{i,c \top} \mathbf{R}(\varphi_t^i)^\top \frac{1}{\beta^2} \mathbf{R}(\varphi_t^i) \Delta \mathbf{y}_t^{i,c} > 1 \end{aligned} \quad (33)$$

and

$$\begin{aligned} \alpha &= a_i + r_c + \epsilon \\ \beta &= b_i + r_c + \kappa \end{aligned} \quad (34)$$

where $\mathcal{K}$ denotes the number of traffic participants that need to be considered in total, $\mathbf{x}_t^E$ and $\mathbf{x}_t^{O_i}$ represent the states of the AV and traffic participant i , $\Delta \mathbf{x}_t^{i,c}, \Delta \mathbf{y}_t^{i,c}$ are the relative

distance between the c -th circle of the AV and the traffic participant i , $\mathbf{R}$ means the rotation matrix depicted in (22), φ_i is the heading angle of traffic participant i , ϵ and κ are the safety margin.

Third, the driving speed v should be limited according to the road regulations. Moreover, the control variables including a and δ should be also limited to reasonable ranges.

$$\begin{aligned} v_{\min} &\leq v_t \leq v_{\max} \\ a_{\min} &\leq a_t \leq a_{\max} \\ \delta_{f\min} &\leq \delta_{ft} \leq \delta_{f\max} \end{aligned} \quad (35)$$

where $a_{\min}$ and $a_{\max}$ are the maximum deceleration and the maximum acceleration, respectively, $\delta_{f\min}$ and $\delta_{f\max}$ are the lower and upper bounds of the steering angle, respectively.

Fourth, the AV also needs to adhere to road boundary constraints. In other words, the lateral deviation distance between the AV and the target driving lane $\mathcal{I}_{opt}(l)$ needs to be restricted within a certain range to prevent the AV from exceeding the road boundaries.

$$d_{left}^{road} \leq \tilde{e}_t^{contour}(\mathcal{I}_{opt}(l), \mathbf{x}_t^E) \leq d_{right}^{road} \quad (36)$$

where d_{left}^{road} and d_{right}^{road} indicate the maximum lateral distance that the AV can deviate from the target driving path $\mathcal{I}_{opt}(l)$ to the left and right boundaries, respectively.

Finally, a fine-level MPC-based motion planner can be constructed by solving the following optimization issue.

$$\begin{aligned} \mathbf{u}_{0:T_{f,mpc}}^* &= \arg \max_{\mathbf{u}^E} \sum_{t=1}^{T_{f,mpc}} \mathcal{J}_{MPC}(\mathbf{x}_t^E, \mathbf{u}_t^E, \mathcal{I}_{opt}) \\ s.t. \quad &\mathbf{x}_{t+1}^E = f(\mathbf{x}_t^E, \mathbf{u}_t^E) \\ &\bigcup_{\substack{c \in [1, 2, 3] \\ i \in [1, 2, \dots, \mathcal{K}]}} cons_t^{i,c}(\mathbf{x}_t^E, \mathbf{x}_t^{O_i}) > 1 \\ &v_{\min} \leq v_t \leq v_{\max} \\ &a_{\min} \leq a_t \leq a_{\max} \\ &\delta_{f\min} \leq \delta_{ft} \leq \delta_{f\max} \\ &d_{left}^{road} \leq \tilde{e}_t^{contour}(\mathcal{I}_{opt}(l), \mathbf{x}_t^E) \leq d_{right}^{road} \end{aligned} \quad (37)$$

where $T_{f,mpc}$ is the prediction horizon of the MPC.

To facilitate the discussion, the hierarchical decision-making method proposed in this paper will be referred to as HDM. Finally, the pseudocode of the proposed HDM is shown in Algorithm 1.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experiment Setup

The experiments in this section mainly include three parts. First, qualitative performance evaluation via the real-world driving dataset is carried out. Second, quantitative performance evaluation is also implemented based on the Carla simulator. Thirdly, the HiL experiment is conducted where the AV is set to interact with surrounding traffic participants in dense traffic and interact with the human driver. Note that all experiments are implemented based on the Carla simulator, and this paper focuses on decision-making techniques in modular autonomous driving systems. Thus, the controlled AV

Algorithm 1 Interactive Decision-Making Integrating Graph Neural Networks and Model Predictive Control

Input: P-GAT with parameters θ , coarse-level policy generator with parameter $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, fine-level MPC planner with parameter $\omega_1, \omega_2, \omega_3, \omega_4, \omega_5$, episodic buffer $\mathcal{D}_e$, empty replay buffer $\mathcal{D}$

for $Episode \leftarrow 1$ to $num_episode$ **do**
 Reset environment state;
 Empty the episodic buffer $\mathcal{D}_e$;
while $Episode$ is not terminal **do**
 Construct driving intention $\mathcal{I}_t$;
 Generate a set of trajectories $\mathcal{T}_t$ of $\mathcal{I}_t$;
for each $\mathcal{T}_t^{v^n} \in \mathcal{T}_t$ **do**
 Query P-GAT with agent history $\mathcal{H}_t$, map $\mathcal{M}_t$, and AV plan $\mathcal{T}_t^{v^n}$ to get prediction results $\mathcal{Y}_t^{v^n}$ based on (26);
end
 Coarse-level policy generator calculates the optimal guidance $\mathcal{I}_{opt}$ according to (31);
 Fine-level MPC planner generates collision constrains (33) with prediction $\mathcal{Y}_t^{v^n}$ and calculates actions with $\mathcal{I}_{opt}$ according to (37);
 Execute action $\mathbf{u}_{0:T_{f,mpc}}^*$ and store agents' states into episodic buffer $\mathcal{D}_e$ every step;
end
 Dump episodic buffer $\mathcal{D}_e$ into replay buffer $\mathcal{D}$;
for $Train_step \leftarrow 1$ to num_step **do**
 Sample a mini-batch $\mathcal{B}$ from $\mathcal{D}$;
 Calculate loss according to ADE metric;
 Update P-GAT parameter θ ;
end
end

first extracts the states of surrounding vehicles directly from the Carla simulator, and the proposed HDM scheme utilizes the states to generate control commands. Then, the control commands are transmitted to the Carla simulator to update environment information.

B. Implementation Details

The overall structure of the P-GAT prediction network is demonstrated in Fig. 2. The proposed approach is trained for 1000 episodes in each urban scenario according to Algorithm 1, and each experiment is conducted multiple times with different random seeds to get test results. In addition, the neural network is trained on a single NVIDIA RTX 3080Ti GPU using Pytorch and Adam optimizer with a learning rate of $2e^{-4}$. To implement qualitative performance evaluation via the real-world driving dataset, the dataset is incorporated into the Carla simulator to construct traffic environment [38]. Furthermore, in the quantitative performance evaluation part, the mixed traffic environment is directly established from the open-source code provided by [29]. Finally, the experiment parameters of HDM and vehicle model are given in Table I.

TABLE I
PARAMETERS OF THE VEHICLE MODEL AND PROPOSED HDM

Parameter	Value	Parameters	Value
l_f	1.81m	l_r	1.33m
T_f	3s	T_h	1s
$T_{f,mpc}$	1.5s	λ_1	10
λ_2	1	λ_3	2
λ_4	0.1	ω_1	50
ω_2	50	ω_3	4
ω_4	6	ω_5	60
$v_{\min}$	0m/s	$v_{\max}$	15m/s
$a_{\min}$	-8m/s ²	$a_{\max}$	2.8m/s ²
$\delta_{f\min}$	-0.22 π rad	$\delta_{f\max}$	0.22 π rad
d_{left}^{road}	-0.5m	d_{right}^{road}	0.5m

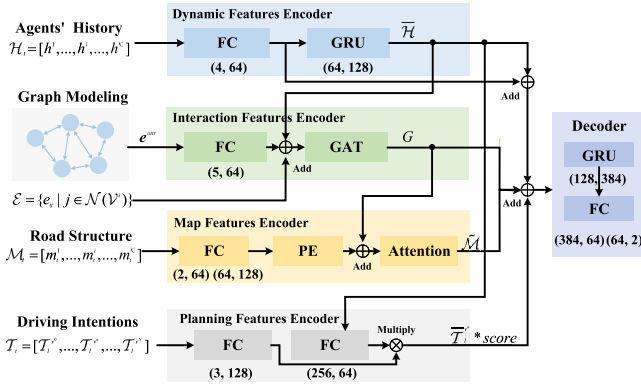


Fig. 2. The overall structure of P-GAT model (note that only key components and their parameters are described, and some tensor operations in the add part are omitted for brevity).

C. Qualitative Performance Evaluation Via Real-World Dataset

To construct complex interaction scenarios, a real-world driving dataset, *i.e.*, openDD is incorporated in the Carla simulator [39]. It focuses on interaction-intensive unregulated roundabouts with varying topologies in Wolfsburg and Ingolstadt, Germany. Qualitative results on four roundabouts in openDD, rdb1, rdb2, rdb3, and rdb6 are demonstrated in Fig. 3. Specifically, Figs. 3(a) and (d) illustrate the decision-making process of the HDM when it encounters a dense traffic flow. When there is no safe passage gap, the HDM opts to decelerate and wait, as shown at 7.3s and 10.0s in Figs. 3(a) and (d). When a sufficiently safe gap appears, the HDM chooses to merge at a moderate speed, as shown at 14.3s and 12.0s in Figs. 3(a) and (d), respectively. Finally, when the driving risks around the AV are low, the HDM accelerates to leave the roundabout, as shown at 21.3s and 16.0s in Figs. 3(a) and (d). Similarly, Fig. 3(b) shows that when a slower vehicle enters in front of the AV, the HDM chooses to decelerate and avoid it. For example, at 6.0s and 11.0s, the AV decelerates to avoid *veh8* and *veh6*, respectively. Fig. 3(c) illustrates that at 1.0s, the AV chooses to enter the roundabout at a speed of 5.76 m/s to avoid *veh2*. At 7.0s, the HDM predicts a collision risk with *veh4* and thus chooses to decelerate and avoid it. Finally, at 11.2s, the HDM decides to exit the roundabout. Note that the results shown in Fig. 3 only include three key moments

TABLE II
ABLATION STUDY OF P-GAT COMPONENTS' IMPACT ON HDM PERFORMANCE

Predictor	Succ.%	Coll.%	TO.%	Pass Time (s)
wo- $\mathcal{H}$	97	3	0	26.2 $\pm$ 0.4
wo- $\tilde{G}$	96	3	1	26.1 $\pm$ 0.4
wo- $\tilde{G}\tilde{\mathcal{M}}$	95	5	0	26.2 $\pm$ 0.1
wo- $\tilde{\mathcal{H}}\tilde{G}\tilde{\mathcal{M}}$	94	6	0	26.2 $\pm$ 0.6
wo- $\tilde{T}$	94	2	4	27.3 $\pm$ 0.4
P-GAT	98	1	1	27.2 $\pm$ 0.4

for each case, and the complete results can be found on the supplementary videos.¹

D. Quantitative Performance Evaluation

To quantitatively evaluate the effectiveness of the proposed HDM method, this section constructs four interactive scenarios in the Carla simulator with a large number of interactive traffic participants, including an unsignalized intersection, a T-junction, a roundabout, and a mixed intersection with both vehicles and pedestrians, as shown in Fig. 4. In addition, the following evaluation metrics are adopted.

- **Success Rate (Succ.%)**: it represents the proportion of test episodes in which the AV safely reaches its destination in time without any collision. A high success rate indicates better driving performance.
- **Collision Rate (Coll.%)**: it represents the proportion of collisions between the AV and surrounding traffic participants. A higher collision rate indicates poorer safety performance.
- **Timeout Rate (TO.%)**: it represents the proportion of instances in which the AV does not complete the driving task within the designated time, without colliding with other traffic participants. A higher timeout rate indicates a lower driving efficiency.
- **Pass Time**: it represents the duration it takes for the AV to complete the driving task. A lower passage time indicates higher driving efficiency.

1) **Ablation Study**: To investigate the impact of different components presented in (25) on the performance of HDM, an ablation study is conducted. Specifically, the roundabout scenario shown in Fig. 4(c) is selected for testing. While other scenarios could also be considered, only the scenario in Fig. 4(c) is explored here due to page limitations. In this study, the key component responsible for capturing interactions in P-GAT is removed to assess its effect on HDM performance. The experimental setup is as follows:

- ① **P-GAT**: HDM utilizes the P-GAT predictor with all components intact.
- ② **wo- $\tilde{T}$** : The planning features shown in (23) and (24) are removed from P-GAT.
- ③ **wo- $\tilde{\mathcal{H}}\tilde{G}\tilde{\mathcal{M}}$** : The interaction features from (14), (18), and (20) are removed from P-GAT.
- ④ **wo- $\tilde{G}\tilde{\mathcal{M}}$** : The interaction features from (18) and (20) are removed from P-GAT.

¹<https://github.com/Kayne0401/Hierarchical-Decision-Making.git>

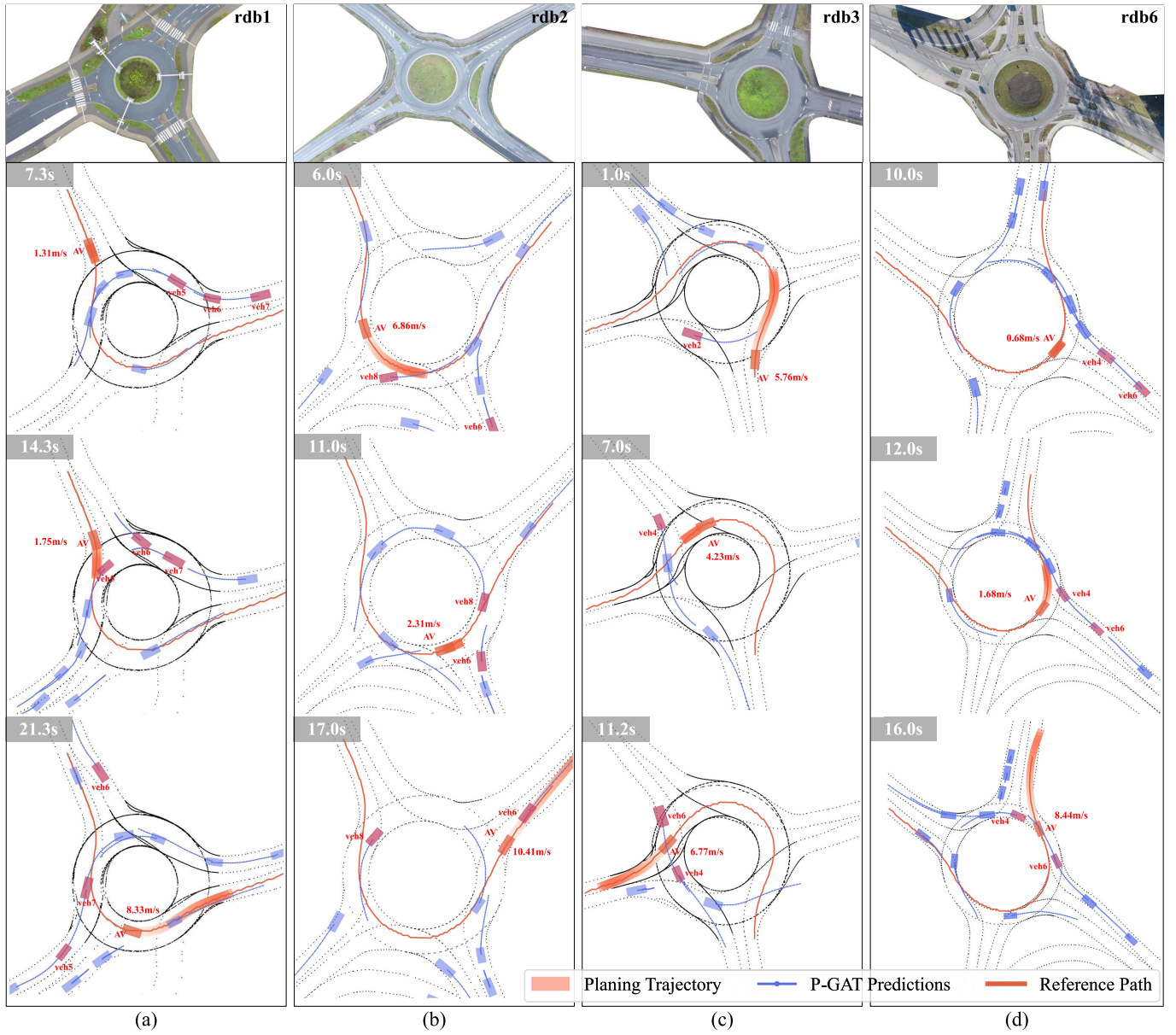


Fig. 3. Qualitative performance evaluation results via openDD dataset.

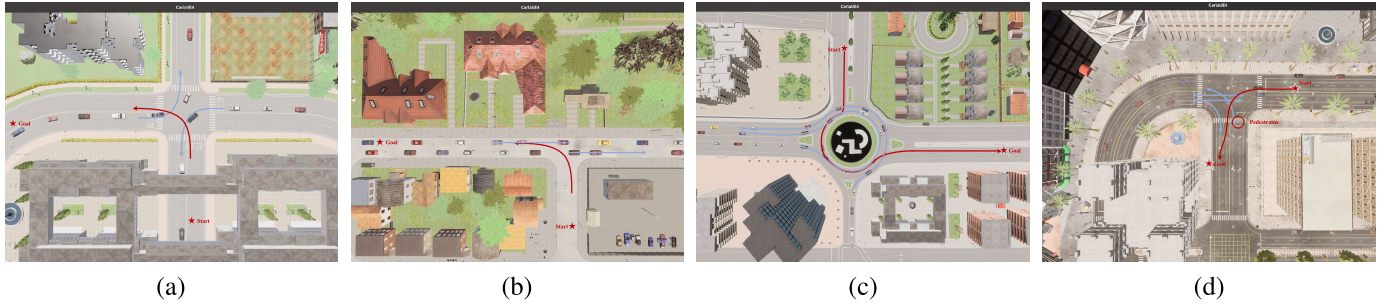


Fig. 4. Test driving scenarios constructed in the CARLA simulator. (a) Unsignalized intersection; (b) T-junction; (c) Roundabout; (d) Mixed Traffic.

- ⑤ $wo-\bar{G}$: The interaction features between the AV and surrounding vehicles, as described in (18), are removed from P-GAT.
- ⑥ $wo-\mathcal{H}$: The historical trajectory features shown in (7) are removed from P-GAT.

Table II presents the results of the ablation study. It is evident that all the interaction features included in (25) contribute to the HDM performance. Notably, the AV plan features extracted in (23) and (24) have the most significant impact. When these features are removed, the success rate of HDM

decreases from 98% to 94%. Additionally, the interaction features $\bar{G}$ between the AV and surrounding vehicles, the road structure features $\bar{M}$, the dynamic features $\bar{H}$, and the historical trajectory features $\mathcal{H}$ all have varying effects on HDM performance.

2) *Comparison Experiments*: This subsection presents several baseline methods to demonstrate the superiority of our proposed approach, as outlined below.

- *Carla Driver Model* [40]: This is the default driver model provided by the Carla simulator, which is the same model used by the surrounding vehicles.
- *GAT-MPC*: This method follows a prediction-then-planning hierarchical approach. The prediction model structure is consistent with P-GAT, but the part related to the AV's intention features is removed. The prediction model is trained using offline collected data. MPC also adopts the same setup as in HDM, but without the interactive guidance.
- *IntMPC* [25]: This state-of-the-art decision-making method combines SAC with a MPC-based planner to navigate through highly interactive driving environments. Notably, the original constant velocity model in IntMPC has been replaced with a Constant Turn Rate and Velocity (CTRV) model to enhance performance.
- *Proximal Policy Optimization (PPO)* [41]: PPO is a widely used reinforcement learning algorithm that optimizes a policy in an on-policy manner. It has been extensively applied to solve autonomous driving decision-making problems.
- *Data-Regularized Q-learning (DrQ)* [42]: This state-of-the-art method for image-based reinforcement learning uses a convolutional neural network (CNN) to encode environment states and employs the SAC algorithm for decision-making.
- *Decision Transformer (DT)* [43]: A strong baseline approach based on the Transformer architecture for reinforcement learning decision-making.
- *Scene-Rep-Transformer (SRT)* [29]: SRT is a state-of-the-art, end-to-end decision-making approach that directly maps environment observations to actions. It augments SAC with Transformer-based scene representation learning to improve interactive decision-making.

The comparative experiments are divided into two parts. The first part compares with existing hierarchical methods, including GAT-MPC and IntMPC, with results shown in Table III. The experimental scenarios in this part include left turns at intersections, left turns at T-junctions, and entering and exiting roundabouts, as shown in Figs. 4(a)-(c). Specifically, in the unsignalized intersection scenario, the Carla Driver Model achieves an 88% success rate, but GAT-MPC (74%) and IntMPC (75%) perform worse, with higher collision rates (GAT-MPC: 25%, IntMPC: 18%). HDM outperforms all, with 98% success, 2% collision rate. In the T-junction scenario, where heavy traffic complicates interactions, GAT-MPC and IntMPC see sharp drops in success (38% and 35%) and high collision rates (GAT-MPC: 60%, IntMPC: 63%). In contrast, HDM maintains the highest success rate (91%), lowest collision rate (9%), and shortest travel time

TABLE III
COMPARISON RESULTS OF POLICY PERFORMANCE

Case	Method	Succ.%	Coll.%	TO.%	Pass Time (s)
(a)	Carla Driver	88	10	2	18.3 ± 0.2
	GAT-MPC	74	25	1	15.8 ± 0.2
	IntMPC	75	18	7	20.4 ± 0.2
	HDM	98	2	0	12.5 ± 0.2
(b)	Carla Driver	60	37	3	21.5 ± 0.2
	GAT-MPC	38	60	1	18.8 ± 0.03
	IntMPC	35	63	2	20.8 ± 0.3
	HDM	91	9	0	18.4 ± 0.2
(c)	Carla Driver	88	12	0	33.5 ± 0.2
	GAT-MPC	67	30	3	26.2 ± 0.3
	IntMPC	91	7	2	30.7 ± 0.2
	HDM	98	1	1	27.2 ± 0.4

TABLE IV
COMPARISON WITH LEARNING-BASED BASELINES
IN MIXED ENVIRONMENT

Method	Succ.%	Coll.%	TO.%	Pass Time (s)
PPO	56	38	6	22.6 ± 1.7
DrQ	84	10	4	21.3 ± 1.4
DT	70	18	8	22.6 ± 1.5
SRT	88	6	2	22.9 ± 1.6
HDM	98	1	1	21.4 ± 0.7

(18.4s). In the roundabout scenario, HDM again excels, with a 98% success rate and the lowest collision rate (1%). IntMPC follows (91% success, 7% collision), while GAT-MPC lags behind (67% success, 30% collision). HDM and IntMPC have similar travel times (27.2s vs. 26.2s). Overall, compared to existing hierarchical methods, HDM better handles high-interaction scenarios, with significant improvements in both safety and driving efficiency.

On the other hand, the second part compares with several existing learning methods including PPO, DrQ, DT and SRT, as presented in Table IV. This evaluation is conducted in a mixed environment with both pedestrian and vehicle traffic, as illustrated in Fig. 4(d). Note that this scenario and comparison algorithms' results are sourced from [29]. From the results, it can be found that HDM stands out with the highest success rate at 98% and the lowest collision rate at 1%, surpassing other methods in safety and reliability. Although DrQ records the shortest passage time at 21.3s, HDM achieves a competitive passage time of 21.4s with higher success and lower collision rates. Furthermore, HDM also demonstrates the lowest timeout rate at 1%, underscoring its efficiency in navigating complex, heterogeneous traffic scenarios. These findings confirm HDM's superior ability to handle interactive and dynamic traffic situations, enhancing both safety and traffic efficiency.

E. Hardware-in-the-Loop Experiments

As depicted in Fig. 5, the HiL test bench includes an Industrial Personal Computer (IPC), NVIDIA Jetson Orin, Speedgoat real-time machine, and Logitech G29 driving simulator. The CARLA simulator, operating on the IPC, handles coarse-level prediction inference. The NVIDIA Orin is tasked with running the fine-level MPC-based planner, while the Speedgoat machine deploys vehicle models. The Logitech

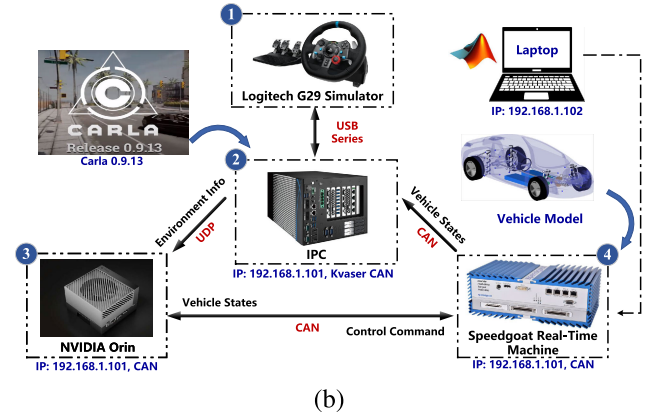
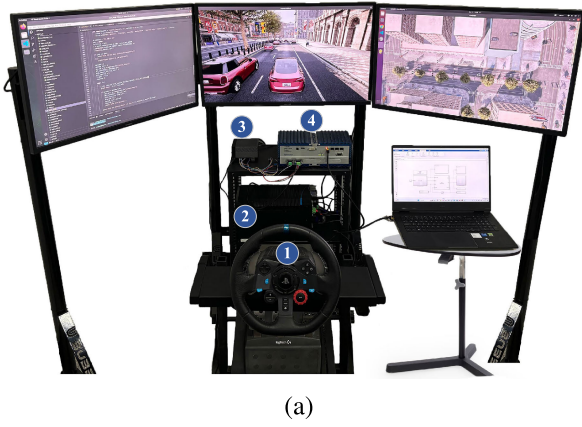


Fig. 5. HiL test bench. (a) HiL platform; (b) Schematic diagram of communication.

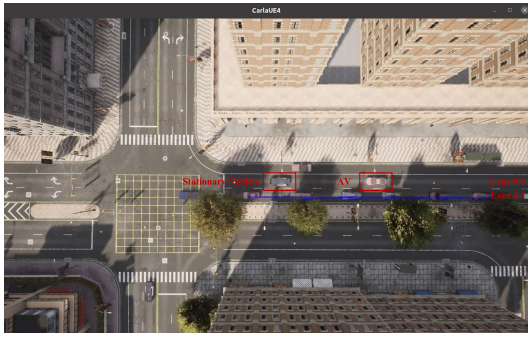


Fig. 6. Schematic diagram of mandatory lane change with dense traffic flow.

G29 is utilized for simulating human-driven vehicles, with the surrounding traffic controlled by the CARLA simulator's driver model.

1) *Mandatory Lane Change in Dense Traffic Flow*: Fig. 6 illustrates a scenario where the AV, located in Lane #0, encounters a stationary vehicle ahead with dense traffic in the adjacent Lane #1. The AV should safely interact with vehicles in Lane #1 to merge in time. Driving speed and steering angle are detailed in Fig. 7. Moreover, the decision-making process of the coarse-level policy generator is shown in Fig. 8(a). Initially, from timestep 0 to 30, the HDM opts to accelerate and maintain its path in Lane #0 due to the distance to the stationary vehicle and dense traffic in Lane #1, which does not offer a safe gap for lane changing. As the AV nears the stationary vehicle between timesteps 30 and 130, the HDM, finding no safe opportunity to change lanes, chooses to decelerate and wait.

At timestep 130, the HDM's P-GAT predicts that upon attempting a lane change, the following vehicle will decelerate to yield, as highlighted in Fig. 7 at key point ①. However, by timestep 135, the following vehicle does not yield as expected, prompting the HDM to cancel the lane change to avoid a collision. It isn't until timestep 275 that the P-GAT predicts sufficient space for a safe lane change, leading the AV to accelerate into Lane #1, as shown at key point ② in Fig. 7, successfully completing the maneuver. Furthermore, Fig. 9 displays P-GAT's predictions under different coarse-level policies at timestep 275. It can be observed that when

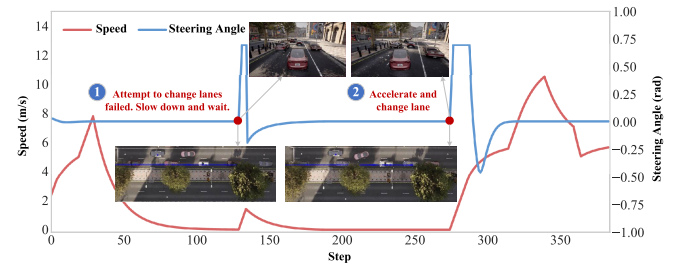


Fig. 7. The driving speed and steering angle of the proposed HDM in the dense traffic case.

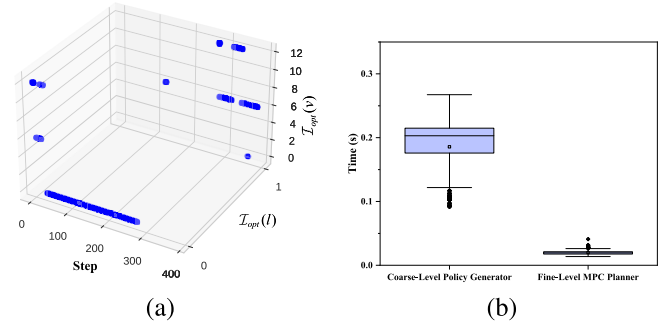


Fig. 8. The decision-making process and real-time performance in the dense traffic case. (a) Decisions of coarse-level policy generator; (b) Real-time performance.

the AV adopts different strategies, P-GAT predicts that the key vehicle (*veh2*) will have different reactions. When the AV maintains its lane (Fig. 9 (a)-(c)), P-GAT foresees that the key vehicle (*veh2*) will not slow down; conversely, when the AV signals a lane change (Fig. 9 (d)-(f)), P-GAT predicts *veh2* will decelerate. The chosen action at timestep 275 according to (31), as seen in Fig. 9 (e), is to change lanes.

Note that the cooperativeness of surrounding vehicles is randomized by adjusting the following distance in the CARLA model, with high traffic density in the target lane. This requires the AV to accurately predict how its actions influence surrounding traffic to ensure they yield safely. Moreover, when P-GAT predictions fail, as demonstrated at time 130 in Fig. 7's key point ①, the fine-level MPC provides real-time planning to mitigate collision risks, halting the lane change.

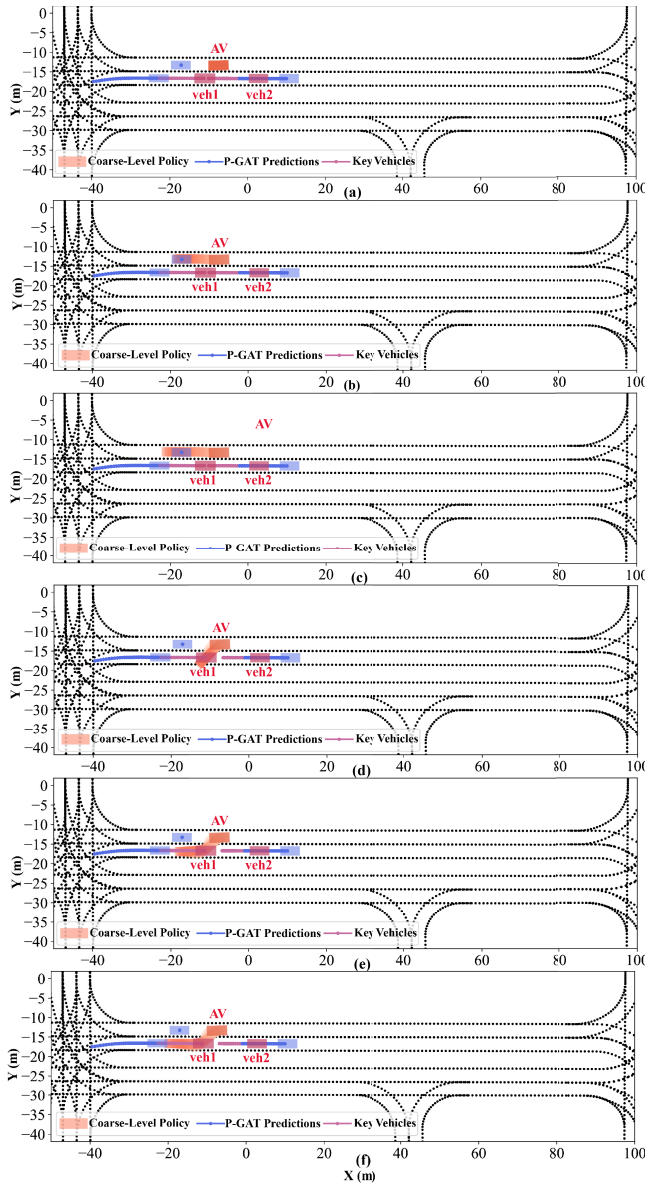


Fig. 9. The prediction results of P-GAT under different coarse-level policies at timestep 275 in the dense traffic case. (a)-(c) demonstrate the prediction results when the AV adopts keeping lane strategy $\mathcal{I}_{opt}(l) = 0$ under $\mathcal{I}_{opt}(v) = 0m/s, 6m/s, 12m/s$, respectively; (d)-(f) demonstrate the prediction results when the AV adopts changing lane strategy $\mathcal{I}_{opt}(l) = 1$ under $\mathcal{I}_{opt}(v) = 0m/s, 6m/s, 12m/s$, respectively. (Note that only the closest five vehicles to the AV are shown in the figures).

The real-time capabilities of the HDM are confirmed in Fig. 8(b), showing that the coarse-level policy generator completes its multiple prediction inferences within 0.2s, meeting the real-time requirement of 2Hz, and the fine-level MPC planner processes within 0.02s, fulfilling the 10Hz requirement. The supplementary videos are provided.²

2) *Interaction With Human Driver*: Fig. 10 depicts the experimental scenario involving interaction between the AV and an aggressively driving human driver, who intentionally obstructs the AV's lane change. The AV's driving speed and steering angle, along with the decision-making process of the coarse-level policy generator, are shown in Figs. 11 and 12.

²<https://github.com/Kayne0401/Hierarchical-Decision-Making.git>

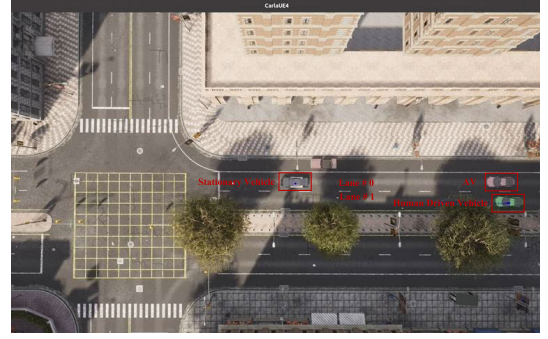


Fig. 10. Schematic diagram of interacting with aggressive human driver.

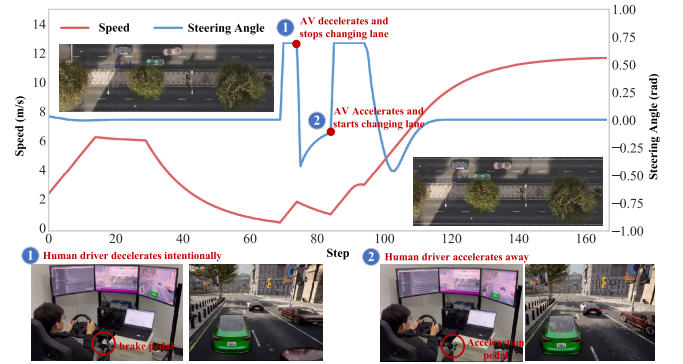


Fig. 11. The driving speed and steering angle of the proposed HDM in the interaction with human driver case.

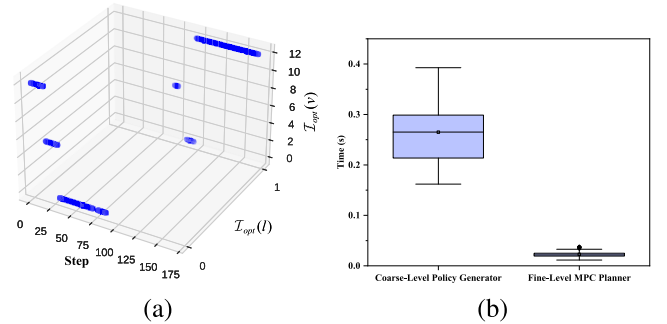


Fig. 12. The decision-making process and real-time performance in the interaction with human driver case. (a) Decisions of coarse-level policy generator; (b) Real-time performance.

Initially, from timestep 0 to 15, with the stationary vehicle ahead at a distance, the HDM maintains a relatively high speed of 12m/s in Lane #0 ($\mathcal{I}_{opt}(l) = 0$ and $\mathcal{I}_{opt}(v) = 12m/s$). As the AV nears the stationary vehicle and encounters the obstructing human-driven vehicle by timestep 15, the HDM opts to reduce speed. By timestep 30, positioned close to the stationary vehicle and still blocked from changing lanes, the AV continues in Lane#0 but decelerates further, waiting for an opportunity to safely change lanes.

At timestep 70, as the human-driven vehicle indicates a departure intent, the HDM's P-GAT assesses a low collision risk, prompting the AV to initiate a lane change to target Lane #1. However, the human driver abruptly decelerates, as illustrated at key point ① in Fig.11. By timestep 75, P-GAT detects a heightened collision risk, leading the HDM to

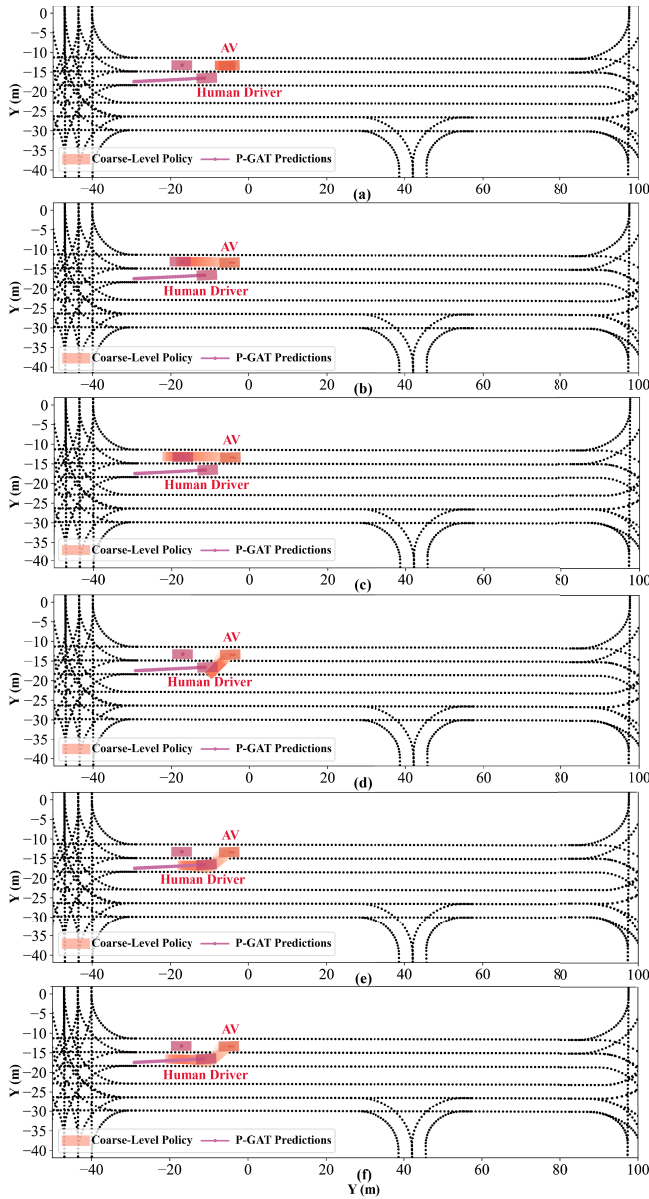


Fig. 13. The prediction results of P-GAT under different coarse-level policies at timestep 85 in the interaction with human driver case. (a)-(c) demonstrate the prediction results when the AV adopts keeping lane strategy $\mathcal{I}_{opt}(l) = 0$ under $\mathcal{I}_{opt}(v) = 0m/s, 6m/s, 12m/s$, respectively; (d)-(f) demonstrate the prediction results when the AV adopts changing lane strategy $\mathcal{I}_{opt}(l) = 1$ under $\mathcal{I}_{opt}(v) = 0m/s, 6m/s, 12m/s$, respectively.

about the lane change. By timestep 85, with the human-driven vehicle accelerating away, the HDM resumes and successfully completes the lane change, shown at key point ② in Fig. 11. Furthermore, Fig. 13 displays the P-GAT's predictions under various coarse-level policies at timestep 85, revealing that the AV's maneuvers do not influence the human-driven vehicle's behavior, regardless of whether the AV maintains its lane ($\mathcal{I}_{opt}(l) = 0$) or attempts a lane change ($\mathcal{I}_{opt}(l) = 1$). This is intuitive because the critical vehicle, the human-driven vehicle, is located in front of the AV at this time. Lastly, the real-time performance of HDM in this scenario is showcased in Fig. 12(b), confirming that the system meets real-time operational requirements.

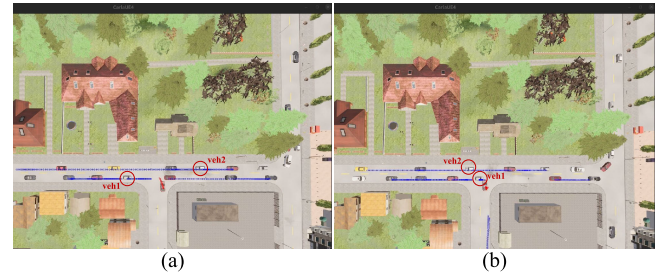


Fig. 14. Failure case of the proposed method, where the AV collides with aggressive vehicles. (a) HDM predicts the aggressive vehicle will yield the AV; (b) collision happens.

F. Failure Case and Limitation Discussion

A typical failure case of the proposed method is illustrated in Fig. 14. Specifically, Fig. 14(a) shows the P-GAT predictions in HDM: if the AV chooses to make a slow left turn, *veh1* and *veh2* are expected to decelerate and yield. However, *veh2* fails to yield, leading to a conflict. As a result, HDM decides to decelerate, but by that time, it has already intersected with the driving path of *veh1*, causing a scraping accident, as shown in Fig. 14(b). The cause of the collision appears to be related to the aggressive behavior of *veh1*, controlled by Carla, which ignores the necessary lateral safety distance. When the AV's path conflicts with that of *veh1*, it fails to stop in time to maintain a safe lateral distance from the AV. This limitation of the proposed model in handling such interactions highlights the need for future research into safe and interaction-aware decision-making.

On the other hand, this paper acknowledges certain limitations in its consideration of computational complexity. While the P-GAT network alleviates the burden by only considering traffic participants within a certain range from the AV, reducing the model parameters and creating a lightweight P-GAT model still require further exploration, especially with limited onboard computing resources. In addition, this paper does not thoroughly explore how the failure risk of the P-GAT model affects the convergence or robustness of the MPC-based planner.

VII. CONCLUSION AND FUTURE WORK

This paper presented a hierarchical decision-making framework, referred to as HDM, tailored for interactive traffic scenarios. Firstly, an interactive motion prediction model, named plan-informed graph attention network (P-GAT) was developed, accounting for the potential impact of the AV's actions on surrounding traffic participants. Subsequently, a coarse-level decision-making policy generator was devised leveraging the P-GAT predictor to provide interaction-aware guidance. Additionally, a fine-level MPC-based planner was introduced, integrating the interaction-aware guidance to generate safe, kinematically feasible trajectories. Finally, the effectiveness of the proposed HDM was verified through qualitative, quantitative and HiL experiments, demonstrating its capability to enhance driving safety and efficiency.

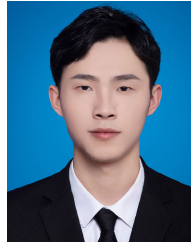
This work does not specifically focus on the impact of various human factors (like fatigue, distraction, etc) on the

interaction between the AV and the human drivers, which are important in real-world driving tasks. In the future, these factors will be investigated to discuss how different human factors might affect the interaction process. More importantly, real-world vehicle experiments will be conducted to further show the applicability of the proposed methods in practical environments.

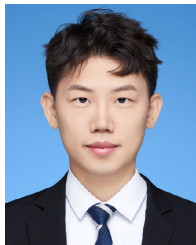
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